

MIAE 221: Materials Science for Engineers

Practice Problem Set #1: Fully Solved & Annotated Master Guide

Question 1: Electron Configurations & Periodic Classification

Problem Statement

Determine whether each of the electron configurations given below is an **inert gas (Group 0)**, a **halogen (Group VIIA)**, an **alkali metal (Group IA)**, an **alkaline earth metal (Group IIA)**, or a **transition metal (Groups IIIB & IIB)**. Justify your choices.

- (a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
- (b) $1s^2 2s^2 2p^6 3s^2 3p^6$
- (c) $1s^2 2s^2 2p^5$
- (d) $1s^2 2s^2 2p^6 3s^2$
- (e) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
- (f) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

Step-by-Step Solutions & Justifications

(a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$

- **Total Electrons:** $2 + 2 + 6 + 2 + 6 + 7 + 2 = 27$ electrons ($Z = 27$, Cobalt, Co).
- **Analysis:** The outermost shell is $n = 4$ with two $4s$ electrons, while the inner $3d$ subshell is partially filled ($3d^7$, holding 7 out of a maximum 10 electrons).
- **Classification:** **Transition Metal (Group VIIIB / d-block).**
- **Justification:** Elements with partially filled inner d -subshells are by definition transition elements.

(b) $1s^2 2s^2 2p^6 3s^2 3p^6$

- **Total Electrons:** $2 + 2 + 6 + 2 + 6 = 18$ electrons ($Z = 18$, Argon, Ar).
- **Analysis:** The outermost principal shell ($n = 3$) has full s and p subshells ($3s^2 3p^6$), comprising a complete stable octet of 8 valence electrons.
- **Classification:** **Inert Gas (Group 0 / Group VIIIA / Noble Gas).**
- **Justification:** A completely filled $s^2 p^6$ valence shell has no unpaired electrons and exceptionally high ionization energy, making it chemically unreactive.

(c) $1s^2 2s^2 2p^5$

- **Total Electrons:** $2 + 2 + 5 = 9$ electrons ($Z = 9$, Fluorine, F).
- **Analysis:** The outermost principal shell ($n = 2$) contains 7 valence electrons ($2s^2 2p^5$), exactly one electron short of a closed octet ($2p^6$).
- **Classification:** **Halogen (Group VIIA).**

- **Justification:** Elements possessing an ns^2np^5 valence structure aggressively seek one additional electron to form a stable -1 anion.

(d) $1s^22s^22p^63s^2$

- **Total Electrons:** $2 + 2 + 6 + 2 = 12$ electrons ($Z = 12$, Magnesium, Mg).
- **Analysis:** The outermost shell is $n = 3$, possessing two valence electrons in the s subshell ($3s^2$).
- **Classification:** **Alkaline Earth Metal (Group IIA).**
- **Justification:** Elements with an ns^2 valence configuration outside a noble gas core readily lose their two outer s -electrons to form $+2$ cations.

(e) $1s^22s^22p^63s^23p^63d^24s^2$

- **Total Electrons:** $2 + 2 + 6 + 2 + 6 + 2 + 2 = 22$ electrons ($Z = 22$, Titanium, Ti).
- **Analysis:** Possesses outer $4s^2$ electrons alongside an incompletely filled inner $3d$ subshell ($3d^2$).
- **Classification:** **Transition Metal (Group IVB / d-block).**
- **Justification:** Contains an unfilled inner d -orbital (d^2), characteristic of transition metals.

(f) $1s^22s^22p^63s^23p^64s^1$

- **Total Electrons:** $2 + 2 + 6 + 2 + 6 + 1 = 19$ electrons ($Z = 19$, Potassium, K).
- **Analysis:** Possesses a single valence electron in the outermost $4s$ subshell ($4s^1$) outside a stable argon core.
- **Classification:** **Alkali Metal (Group IA).**
- **Justification:** Elements with an ns^1 valence structure have exceptionally low first ionization energies and form $+1$ cations.

Question 2: Bonding Differences & Percent Ionic Character

Part I: Fundamental Differences Among Primary Bonds

1. Ionic Bonding:

- *Mechanism:* Mutual electrostatic attraction between positively charged metal cations and negatively charged nonmetal anions following complete electron transfer.
- *Directionality:* **Non-directional** (Coulombic fields act uniformly in all spherical directions).
- *Characteristics:* High melting points, brittle, electrical insulators as solids, conductive when molten.

2. Covalent Bonding:

- *Mechanism:* Localized sharing of valence electron pairs between adjacent atoms of similar electronegativities to fill atomic orbitals.
- *Directionality:* **Highly directional** (forms along specific hybrid orbital angles, e.g., 109.5° for sp^3).
- *Characteristics:* High hardness, directional cleavage, wide range of electrical conductivity.

3. Metallic Bonding:

- *Mechanism:* Positive ion cores held together by electrostatic attraction to a shared, delocalized "sea of mobile conduction electrons."
- *Directionality:* **Non-directional** (electron cloud flows freely around ion cores).
- *Characteristics:* High electrical and thermal conductivity, excellent ductility and plastic malleability.

Part II: Compound Classification & % Ionic Character

Pauling Electronegativity Reference Values: $H = 2.1$, $C = 2.5$, $N = 3.0$, $O = 3.5$, $F = 4.0$, $Na = 0.9$, $Cl = 3.0$, $Ca = 1.0$, $Ti = 1.5$, $Zn = 1.6$, $Te = 2.1$

Pauling Formula:

Compound	Dominant Bond Type	$\Delta X = X_A - X_B $	% Ionic Character
a) Brass (Cu-Zn)	Metallic	N/A	N/A
b) TiO₂	Mixed (Predom. Ionic)	$ 3.5 - 1.5 = 2.0$	$[1 - e^{-0.25(4.0)}] = \mathbf{63.2\%}$
c) Diamond (C)	Pure Covalent	$ 2.5 - 2.5 = 0.0$	$[1 - e^{-0.25(0)}] = \mathbf{0.0\%}$
d) ZnTe	Mixed (Predom. Covalent)	$ 2.1 - 1.6 = 0.5$	$[1 - e^{-0.25(0.25)}] = \mathbf{6.1\%}$
e) Rubber	Covalent & van der Waals	Chain: Covalent; Interchain: Secondary (crosslinks are covalent)	N/A
f) H₂O (within molecule)	Polar Covalent	$ 3.5 - 2.1 = 1.4$	$[1 - e^{-0.25(1.96)}] = \mathbf{38.7\%}$
g) H₂O (between molecules)	Hydrogen Bonding	Intermolecular dipole	N/A
h) Solid Xenon	van der Waals	Fluctuating induced dipole	N/A
i) Bronze (Cu-Sn)	Metallic	N/A	N/A
j) Nylon	Covalent & Hydrogen	Chain: Covalent; Interchain: Hydrogen bonds	N/A
k) CaCl₂	Predominantly Ionic	$ 3.0 - 1.0 = 2.0$	$[1 - e^{-0.25(4.0)}] = \mathbf{63.2\%}$

Question 3: $K^+ - Cl^-$ Potential Energy Curve & Equilibrium State

Problem Statement

For a $K^+ - Cl^-$ ion pair, attractive and repulsive energies depend on interionic distance r according to:

$$E_A(r) = -\frac{1.436}{r} \quad E_R(r) = \frac{5.8 \times 10^{-6}}{r^9}$$

Where energies are in eV and r is in nanometers (nm). The net potential energy is $E_N(r) = E_A(r) + E_R(r)$.

- (a)** Calculate the equilibrium interionic spacing (r_0).
- (b)** Calculate the bonding energy (E_0).

Step-by-Step Mathematical Derivation

Part (a): Equilibrium Spacing r_0

At equilibrium, the net interatomic force is zero, meaning the net potential energy curve reaches its minimum:

$$\left. \frac{dE_N}{dr} \right|_{r=r_0} = 0$$

Given:

$$E_N(r) = -1.436r^{-1} + 5.8 \times 10^{-6}r^{-9}$$

Differentiating with respect to r :

$$\frac{dE_N}{dr} = (-1.436)(-1)r^{-2} + (5.8 \times 10^{-6})(-9)r^{-10} = \frac{1.436}{r^2} - \frac{5.22 \times 10^{-5}}{r^{10}}$$

Setting the derivative to zero at $r = r_0$:

$$\frac{1.436}{r_0^2} = \frac{5.22 \times 10^{-5}}{r_0^{10}}$$

Cross-multiplying:

$$1.436 r_0^8 = 5.22 \times 10^{-5} \implies r_0^8 = \frac{5.22 \times 10^{-5}}{1.436} = 3.6351 \times 10^{-5}$$

Taking the 8th root:

$$r_0 = (3.6351 \times 10^{-5})^{\frac{1}{8}} = \mathbf{0.2789 \text{ nm}} \approx \mathbf{0.279 \text{ nm}} \quad (2.79 \text{ \AA})$$

Part (b): Bonding Energy E_0

Substitute $r_0 = 0.2789 \text{ nm}$ into $E_N(r)$:

$$E_0 = E_N(r_0) = -\frac{1.436}{0.2789} + \frac{5.8 \times 10^{-6}}{(0.2789)^9}$$

Evaluating each term:

- Attractive Energy: $E_A = -\frac{1.436}{0.2789} = -5.1488 \text{ eV}$
- Repulsive Energy: $E_R = \frac{5.8 \times 10^{-6}}{(0.2789)^9} = \frac{5.8 \times 10^{-6}}{1.0138 \times 10^{-5}} = +0.5721 \text{ eV}$

Summing the two terms:

$$E_0 = -5.1488 + 0.5721 = -4.577 \text{ eV} \approx -4.58 \text{ eV}$$

(Alternative algebraic check: $E_0 = -\frac{A}{r_0}(1 - 1/n) = -\frac{1.436}{0.2789}(1 - \frac{1}{9}) = -5.1488 \times \frac{8}{9} = -4.577 \text{ eV}$. Perfect match!)

Question 4: Electrostatic Force Balance Between K^+ and O^{2-}

Problem Statement

- **(a)** Calculate the force of attraction using Coulomb's law between a K^+ ion ($z_1 = +1$) and an O^{2-} ion ($z_2 = -2$) separated by $r = 1.5 \text{ nm}$.
- **(b)** Calculate the repulsive force if $r = 1.5 \text{ nm}$ is the equilibrium interatomic spacing.

Step-by-Step Solution

Part (a): Force of Attraction (F_A)

Coulomb's Law for interionic attractive force:

$$F_A = \frac{|z_1||z_2|e^2}{4\pi\epsilon_0 r^2}$$

Physical Constants:

- Electron charge: $e = 1.602 \times 10^{-19} \text{ C}$
- Permittivity of free space: $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2) \text{ [or F/m]}$

- Valences: $|z_1| = 1 (K^+)$, $|z_2| = 2 (O^{2-}) \implies |z_1 z_2| = 2$
- Distance: $r = 1.5 \text{ nm} = 1.5 \times 10^{-9} \text{ m}$

Substitute values into the equation:

$$F_A = \frac{2 \times (1.602 \times 10^{-19})^2}{4\pi \times (8.854 \times 10^{-12}) \times (1.5 \times 10^{-9})^2}$$

Calculate numerator:

$$\text{Numerator} = 2 \times 2.5664 \times 10^{-38} = 5.1328 \times 10^{-38} \text{ C}^2$$

Calculate denominator:

$$\text{Denominator} = 4\pi \times 8.854 \times 10^{-12} \times 2.25 \times 10^{-18} = 1.11265 \times 10^{-10} \times 2.25 \times 10^{-18} = 2.5035 \times 10^{-28}$$

Divide:

$$F_A = \frac{5.1328 \times 10^{-38}}{2.5035 \times 10^{-28}} = 2.05 \times 10^{-10} \text{ N} \quad (0.205 \text{ nN})$$

Part (b): Repulsive Force at Equilibrium (F_R)

By definition, at the equilibrium spacing $r_0 = 1.5 \text{ nm}$, the net interatomic force must be zero:

$$F_N = F_A + F_R = 0 \implies F_R = -F_A$$

Since the attractive force acts inward with magnitude $2.05 \times 10^{-10} \text{ N}$, the repulsive force must act outward with equal magnitude:

$$F_R = +2.05 \times 10^{-10} \text{ N} \quad (+0.205 \text{ nN})$$

Question 5: Macroscopic to Microscopic Stoichiometric Calculations

Part (a): Gold Wire Atom Count

Problem: Calculate the number of atoms in a gold wire that is 0.70 mm in diameter and 8.0 cm in length. The density of gold is $\rho = 19.3 \text{ g/cm}^3$.

Step 1: Geometric Volume Calculation

- Diameter: $d = 0.70 \text{ mm} = 0.070 \text{ cm}$
- Length: $L = 8.0 \text{ cm}$
- Cross-sectional area:

$$A_{\text{cross}} = \frac{\pi}{4} d^2 = \frac{\pi}{4} (0.070 \text{ cm})^2 = 3.8485 \times 10^{-3} \text{ cm}^2$$

- Total volume:

$$V = A_{\text{cross}} \times L = 3.8485 \times 10^{-3} \times 8.0 = 0.030788 \text{ cm}^3$$

Step 2: Mass Calculation

$$m = \rho \times V = 19.3 \text{ g/cm}^3 \times 0.030788 \text{ cm}^3 = 0.5942 \text{ g}$$

Step 3: Moles of Gold

From the periodic table, atomic weight of Gold (Au): $A_{Au} = 196.97 \text{ g/mol}$.

$$n_{mol} = \frac{m}{A_{Au}} = \frac{0.5942 \text{ g}}{196.97 \text{ g/mol}} = 3.0167 \times 10^{-3} \text{ mol}$$

Step 4: Total Number of Atoms

Using Avogadro's number ($N_A = 6.022 \times 10^{23} \text{ atoms/mol}$):

$$N = n_{mol} \times N_A = 3.0167 \times 10^{-3} \times 6.022 \times 10^{23} = \mathbf{1.82 \times 10^{21} \text{ atoms}}$$

Part (b): Number of Atoms in One Ton of Iron

Problem: Calculate the number of atoms in one ton of iron (Fe). The atomic weight of iron is $A_{Fe} = 55.85 \text{ g/mol}$.

1. Metric Ton Calculation (1 metric ton = 1000 kg = 10^6 g):

$$m = 1.0 \times 10^6 \text{ g}$$

$$n_{mol} = \frac{m}{A_{Fe}} = \frac{1.0 \times 10^6 \text{ g}}{55.85 \text{ g/mol}} = 17,905.1 \text{ mol}$$

$$N = n_{mol} \times N_A = 17,905.1 \times 6.022 \times 10^{23} = \mathbf{1.08 \times 10^{28} \text{ atoms}}$$

2. US Short Ton Calculation (1 short ton = 2000 lb = 907.185 kg = $9.07185 \times 10^5 \text{ g}$):

$$n_{mol} = \frac{9.07185 \times 10^5 \text{ g}}{55.85 \text{ g/mol}} = 16,243.2 \text{ mol}$$

$$N = 16,243.2 \times 6.022 \times 10^{23} = \mathbf{9.78 \times 10^{27} \text{ atoms}}$$